

Introduction:

The **self-similar aftershock rates (SSAR)** model proposed in *Davidson and Baiesi (2016)* is a model of earthquake–earthquake triggering, similar to the ETAS model

Methods:

In the ordinary framework of Omori-Utsu law, the triggering ability/productivity is expressed as:

$$K = K_0 10^{\alpha m'}$$

while considering the aftershocks following G-R relation, so that the number of triggered events of magnitude m_{as} scales:

$$10^{\alpha m' - b_{as} m_{as}} = 10^{(\alpha - b_{as})m' + b_{as} \Delta m}, \quad \Delta m = m' - m_{as}$$

Thus, the triggering rate depends not only on the magnitude difference Δm and also explicitly depends on the magnitude of the mainshock m'

The SSAR model recasts the standard Omori–Utsu rate equation into a **self-similar** version: (only depends on Δm)

$$r(m_{as}, t | m', 0) = \frac{1}{\tau_{\Delta m}} \left(\frac{t}{c_{\Delta m}} + 1 \right)^{-p},$$

with time scales:

$$c_{\Delta m} = c_0 10^{g \Delta m} \text{ and } \tau_{\Delta m} = \tau_0 10^{-z \Delta m},$$

where g and z are universal scaling exponents, c_0 and τ_0 are constant prefactors, and $p \geq 1$

Thus, the magnitude difference controls not only "intensity of triggering" but also "timescale of triggering"

The total number of triggered events of magnitude m_{as} for a given trigger m' is:

$$N(m_{as} | m') = \int_0^\infty \frac{1}{\tau_{\Delta m}} \left(\frac{t}{c_{\Delta m}} + 1 \right)^{-p} dt = \frac{c_{\Delta m}}{\tau_{\Delta m} (p - 1)},$$

Notice: the substitution method is used here:

Let: $u = \left(\frac{t}{c_{\Delta m}} + 1 \right)$

so that:

$$dt = c_{\Delta m} du$$

when $t = 0, u = 1$ and $t = \infty, u = \infty$

so:

$$N(m_{as} | m') = \frac{c_{\Delta m}}{\tau_{\Delta m}} \int_1^\infty u^{-p} du = \frac{c_{\Delta m}}{\tau_{\Delta m} (p - 1)}$$

Then substitute the known variables:

$$N(m_{as} | m') = \frac{c_0}{\tau_0 (p - 1)} 10^{(g+z) \Delta m}$$

and integrating it from cut-off magnitude to infinity:

$$N_{>}(m_{\text{cut}} | m') = \frac{c_0 10^{(g+z)(m' - m_{\text{cut}})}}{\tau_0 (p - 1) (g + z) \ln 10},$$

where the scaling relation is (*Shcherbakov et al., 2004*):

$$b_{as} = g + z$$

The generalized productivity relation is:

$$K_{\Delta m} = K_0 10^{\alpha \Delta m} \\ \text{with } K_0 = c_0^p / \tau_0$$

Then:

$$\alpha = b_{as} + g(p - 1)$$

The conditional intensity function can be expressed as:

$$\lambda(t, m, \mathbf{r}) = \mu s_0(m) + \sum_{t' < t} \kappa(m') \psi_{m, m'}(t - t')$$

$$s(m) \zeta_{m'}(\mathbf{r} - \mathbf{r}'),$$

where:

$$\begin{cases} s_0(m) = \beta e^{-\beta(m-m_{\text{cut}})} \\ s(m) = \beta_{as} e^{-\beta_{as}(m-m_{\text{cut}})}, \\ \kappa(m') = A e^{\beta_{as}(m'-m_{\text{cut}})}, \\ \psi_{m, m'}(t) = \frac{(p-1) c_{\Delta m}^{p-1}}{(t + c_{\Delta m})^p}. \end{cases}$$

Three conditions to verify the magnitude correlation:

$P(\cdot)$ refers to the cumulative distribution function (CDF) of the ordered and randomized catalogs

Case 1: uncondition

$$\delta P(m_0) = P(\Delta m < m_0) - P(\Delta m^* < m_0)$$

Case 2: Time restraint

$$\delta P(m_0 | \Delta t < y) = P(\Delta m < m_0 | \Delta t < y) - P(\Delta m^* < m_0 | \Delta t < y)$$

Case 3: Time restraint and Mother-daughter pairs

$$\delta P(m_0 | \Delta t < y \& M-D) = P(\Delta m < m_0 | \Delta t < y \& M-D) - P(\Delta m^* < m_0 | \Delta t < y \& M-D)$$

and there are two ways to choose the magnitudes m^* in the case of 2 or case 3

1. sub-catalog randomizing
2. full-catalog randomizing